In [1]:

```python
import pandas as pd
from matplotlib import pyplot as plt
import numpy as np
import seaborn as sns
#laoding the data using pandas
df = pd.read_csv('time_series.csv')
#viewing the first 5 rows of the data to see how it looks
df.head(5)
```

Out[1]:

| | date | X1 | X2 | X3 | X4 | X5 | X6 | Y |
|---|---|---|---|---|---|---|---|---|
| **0** | 2016-07-01 00:00:00 | 5.827 | 2.009 | 1.599 | 0.462 | 4.203 | 1.340 | 30.531000 |
| **1** | 2016-07-01 01:00:00 | 5.693 | 2.076 | 1.492 | 0.426 | 4.142 | 1.371 | 27.787001 |
| **2** | 2016-07-01 02:00:00 | 5.157 | 1.741 | 1.279 | 0.355 | 3.777 | 1.218 | 27.787001 |
| **3** | 2016-07-01 03:00:00 | 5.090 | 1.942 | 1.279 | 0.391 | 3.807 | 1.279 | 25.044001 |
| **4** | 2016-07-01 04:00:00 | 5.358 | 1.942 | 1.492 | 0.462 | 3.868 | 1.279 | 21.948000 |

In [2]:

```python
#checking the data types of all columns to make sure that correct formats are used
df.dtypes
```

Out[2]:

```
date     object
X1      float64
X2      float64
X3      float64
X4      float64
X5      float64
X6      float64
Y       float64
dtype: object
```

In [3]:

```python
#converting date column into datetime format
df['date'] = pd.to_datetime(df['date'])
#viewing the updated date format to see if it has worked
df.dtypes
```

Out[3]:

```
date    datetime64[ns]
X1             float64
X2             float64
X3             float64
X4             float64
X5             float64
X6             float64
Y              float64
dtype: object
```

In [4]:

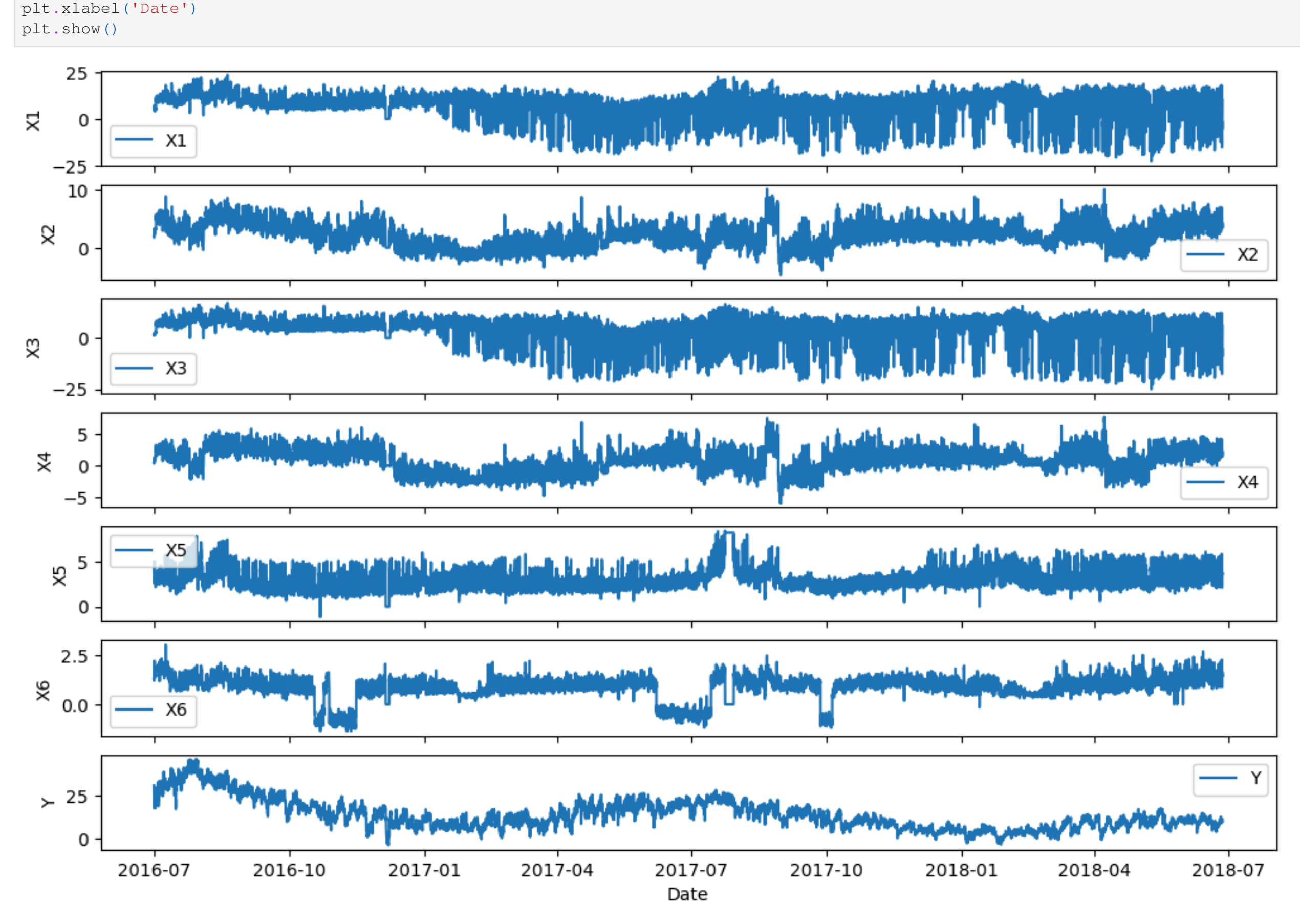
```python
#making date column become the index of the table. This makes it easire to work with the data
df.set_index('date', inplace=True)
df.head(5)
```

Out[4]:

| | X1 | X2 | X3 | X4 | X5 | X6 | Y |
|---|---|---|---|---|---|---|---|
| **date** | | | | | | | |
| **2016-07-01 00:00:00** | 5.827 | 2.009 | 1.599 | 0.462 | 4.203 | 1.340 | 30.531000 |
| **2016-07-01 01:00:00** | 5.693 | 2.076 | 1.492 | 0.426 | 4.142 | 1.371 | 27.787001 |
| **2016-07-01 02:00:00** | 5.157 | 1.741 | 1.279 | 0.355 | 3.777 | 1.218 | 27.787001 |
| **2016-07-01 03:00:00** | 5.090 | 1.942 | 1.279 | 0.391 | 3.807 | 1.279 | 25.044001 |
| **2016-07-01 04:00:00** | 5.358 | 1.942 | 1.492 | 0.462 | 3.868 | 1.279 | 21.948000 |

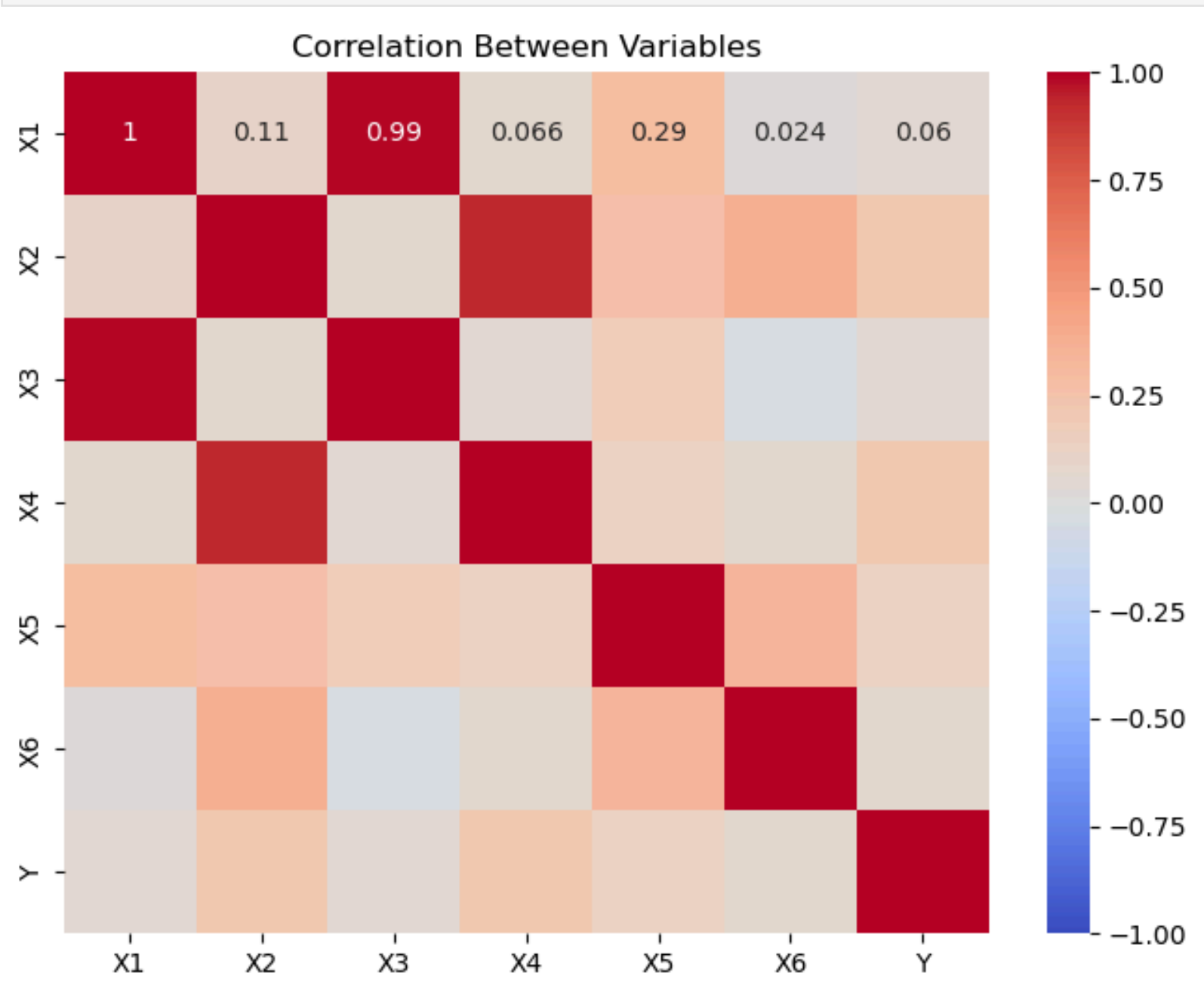
In [9]:

```python
#plotting each variable to see how it is distributed
#ceating a subplot
fig, axes = plt.subplots(nrows=len(df.columns), ncols=1, figsize=(12, 8), sharex=True)
#looping through each column and plotting the data
for i, col in enumerate(df.columns):
    axes[i].plot(df.index, df[col], label=col)
    axes[i].set_ylabel(col)
    axes[i].legend()

plt.xlabel('Date')
plt.show()
```



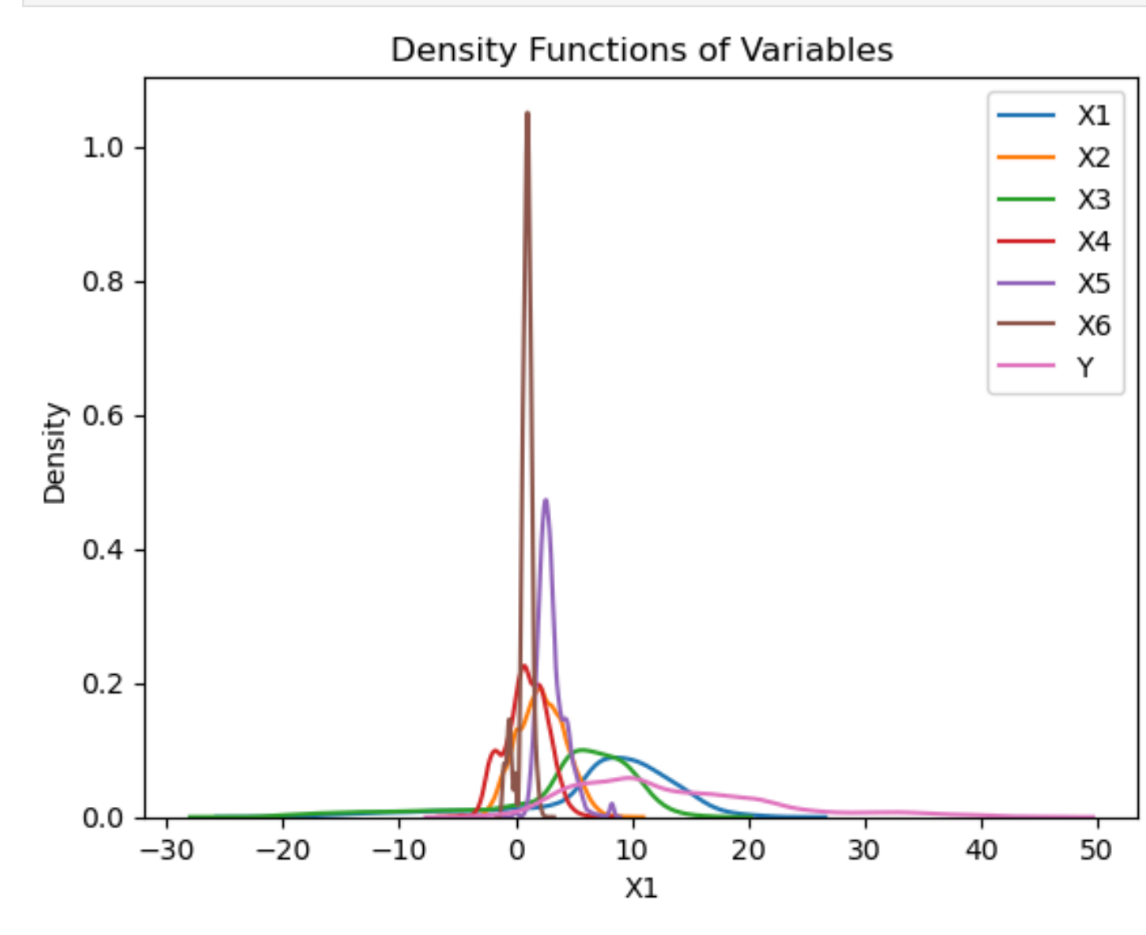
In [10]:

```python
# Correlation heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(df.corr(), annot=True, cmap="coolwarm", vmin=-1, vmax=1)
plt.title("Correlation Between Variables")
plt.show()
```



In [11]:

```python
# Density plots
for col in df.columns:
    sns.kdeplot(df[col], label=col)
plt.title("Density Functions of Variables")
plt.legend()
plt.show()
```



In [12]:

```python
#splitting the data
train_data = df['2016-07-01' : '2017-06-26']
validation_data = df['2017-06-27':'2017-10-24']
test_data = df['2017-10-25': '2018-02-21']
```

In [13]:

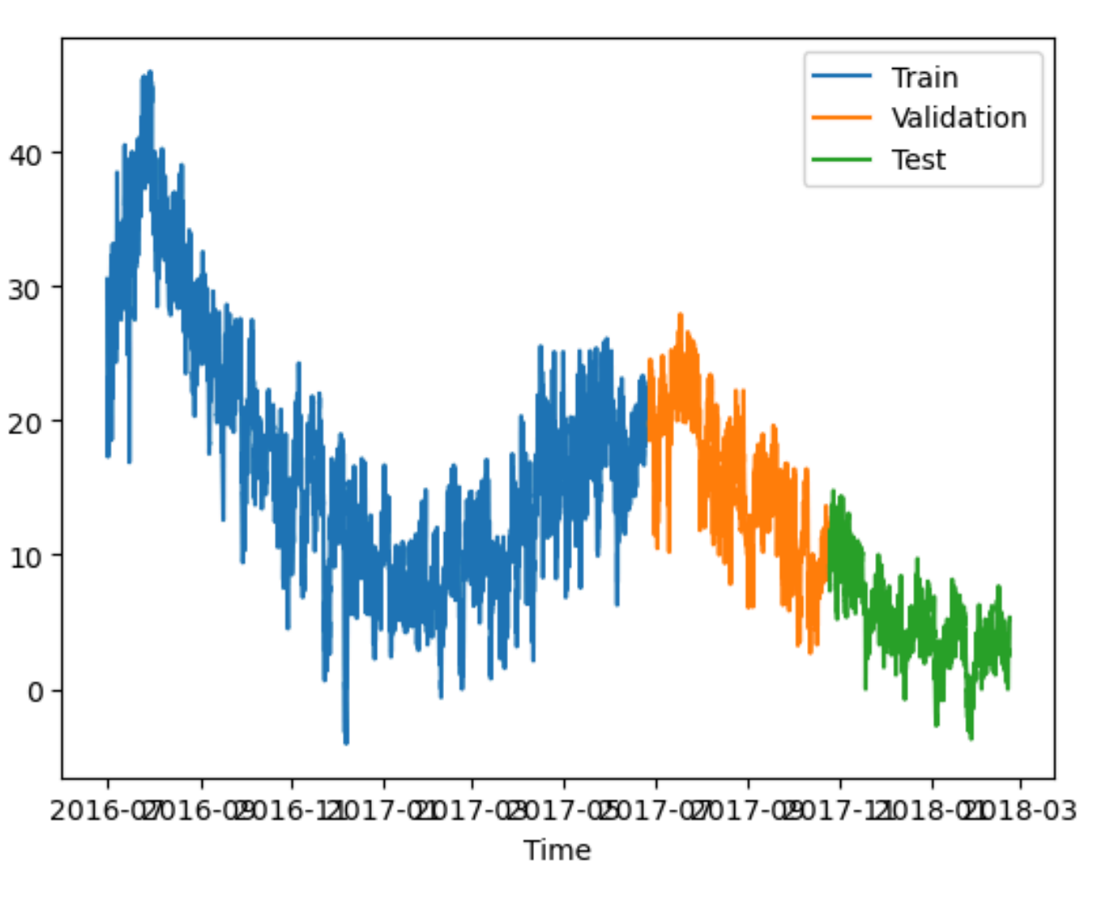
```python
train_y = train_data['Y']
validation_y = validation_data['Y']
test_y = test_data['Y']
```

In [16]:

```python
plt.plot(train_y, label= 'Train')
plt.plot(validation_y, label= 'Validation')
plt.plot(test_y, label= 'Test')
plt.legend()
plt.xlabel('Time')
```

Out[16]:

```
Text(0.5, 0, 'Time')
```



In [35]:

```python
# creating a scaler function

def standard_scaler(x_data, X_column):
    mean = np.mean(X_column)
    std_dev = np.std(X_column)
    z = (x_data - mean) / std_dev
    return z
```

In [36]:

```python
#creating a new data empty datafarne to store the standardized data
train_data_standardized = pd.DataFrame()

for column in train_data.columns:
    train_data_standardized[column] = standard_scaler(train_data[column].values ,train_data[column] )
train_data_standardized
```

Out[36]:

| | X1 | X2 | X3 | X4 | X5 | X6 | Y |
|---|---|---|---|---|---|---|---|
| **0** | -0.362731 | -0.005914 | -0.630120 | -0.149847 | 1.389276 | 0.876988 | 1.461431 |
| **1** | -0.385794 | 0.026178 | -0.649512 | -0.168539 | 1.329613 | 0.925968 | 1.162039 |
| **2** | -0.478047 | -0.134281 | -0.688116 | -0.205403 | 0.972615 | 0.684229 | 1.162039 |
| **3** | -0.489578 | -0.038006 | -0.688116 | -0.186711 | 1.001957 | 0.780609 | 0.862756 |
| **4** | -0.443452 | -0.038006 | -0.649512 | -0.149847 | 1.061620 | 0.780609 | 0.524958 |
| **...** | ... | ... | ... | ... | ... | ... | ... |
| **8659** | 0.271335 | -0.326354 | 0.258674 | 0.496055 | 0.465971 | -2.155028 | 0.256335 |
| **8660** | 0.490436 | -0.390538 | 0.413270 | 0.256178 | 1.032277 | -2.155028 | 0.310125 |
| **8661** | 0.317462 | -0.486814 | 0.310146 | 0.532919 | 0.197977 | -2.202428 | 0.340784 |
| **8662** | 0.248272 | -0.262170 | 0.265017 | 0.329907 | 0.286982 | -2.155028 | 0.317763 |
| **8663** | 0.363588 | 0.026178 | 0.368142 | 0.846525 | 0.108972 | -1.865889 | 0.348422 |

8664 rows × 7 columns

In [37]:

```python
#checking if standardization has worked
print("Train mean (Y):", train_data_standardized['Y'].mean(), "std:", train_data_standardized['Y'].std())
```

```
Train mean (Y): 1.4795669562822536e-15 std: 1.0000577150607906
```

In [38]:

```python
validation_data_standardized = pd.DataFrame()

for column in validation_data.columns:
    validation_data_standardized[column] = standard_scaler(validation_data[column].values ,validation_data[column] )
```

In [39]:

```python
test_data_standardized = pd.DataFrame()

for column in test_data.columns:
    test_data_standardized[column] = standard_scaler(test_data[column].values ,test_data[column] )
```

In [ ]:

In [40]:

```python
#Implement matrix multiplication and inversion using numpy.
#selecting our matrix X and vector y from the train data

def LSE(X, y):
    # Concatenate a column of ones to X for the intercept term
    X_with_bias = np.c_[np.ones(X.shape[0]), X]  # shape (n_samples, n_features + 1)
    X_transpose = X_with_bias.T  # calculating x transpose
    beta = np.linalg.inv(X_transpose @ X_with_bias) @ (X_transpose @ y)
    return beta


X = train_data_standardized.iloc[:, : -1].values
y = train_data_standardized.iloc[:, -1].values


#Compute the estimated parameters using the LSE method.
beta = LSE(X, y)
print("Estimated parameters:", beta)
```

```
Estimated parameters: [ 5.29349411e-16 -3.15373366e-04  2.25661046e-01  1.46149538e-02
  3.10183880e-01  1.80987385e-01  1.47761850e-01]
```

In [53]:

```python
# Predict on train, validation, and test sets
def predic_function(X, beta):
    X_with_bias = np.c_[np.ones(X.shape[0]), X]
    y_pred = X_with_bias @ beta
    return y_pred
#separeting label and variables
X_train = train_data_standardized.iloc[:, : -1].values
y_train = train_data_standardized.iloc[:, -1: ].values

X_validation = validation_data_standardized.iloc[:, : -1].values
y_validation = validation_data_standardized.iloc[:, -1: ].values

X_test = test_data_standardized.iloc[:, : -1].values
y_test = test_data_standardized.iloc[:,-1 : ].values

# predicting y for each section of the data
y_pred_train = predic_function(X_train, beta)
y_pred_validation = predic_function(X_validation, beta)
y_pred_test = predic_function(X_test, beta)
```

In [54]:

```python
#Mean Squared Error (MSE)
def MSE(y, y_hat):
    mse_loss = np.mean((y - y_hat) ** 2)
    return mse_loss

#Root Mean Squared Error (RMSE)
def RMSE(y, y_hat):
    rmse_loss = np.sqrt(MSE(y, y_hat))
    return rmse_loss
```

In [55]:

```python
train_MSE = MSE(y_train, y_pred_train)
train_RMSE = RMSE(y_train, y_pred_train)
print("Mean Squared Error:", train_MSE)
print("Root Mean Squared Error:", train_RMSE)
```

```
Mean Squared Error: 1.3987402890649085
Root Mean Squared Error: 1.182683511792106
```

In [56]:

```python
validation_MSE = MSE(y_validation, y_pred_validation)
validation_RMSE = RMSE(y_validation, y_pred_validation)
print("Mean Squared Error:", validation_MSE)
print("Root Mean Squared Error:", validation_RMSE)
```

```
Mean Squared Error: 1.4036150351273549
Root Mean Squared Error: 1.184742602900459
```

In [57]:

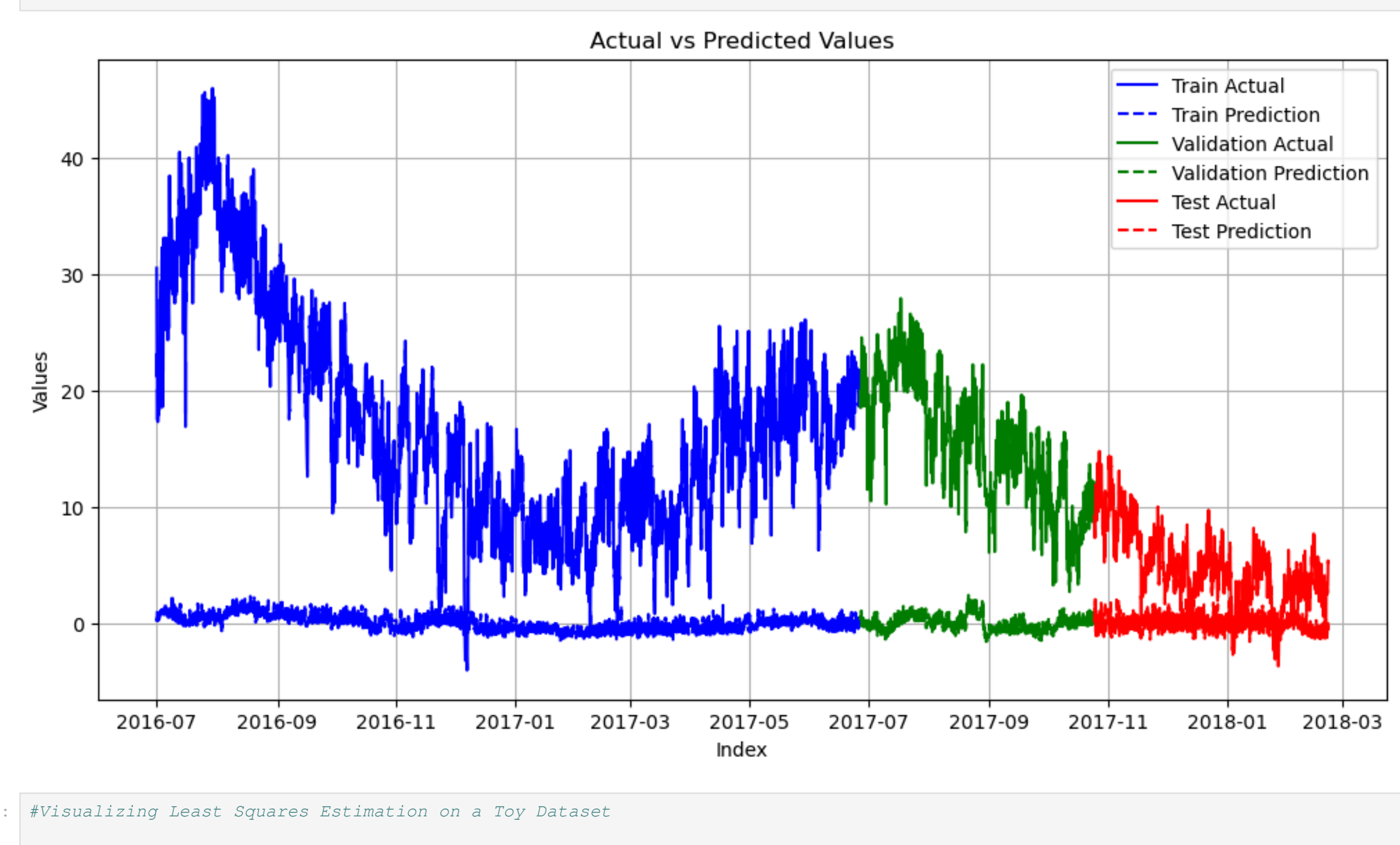
```python
test_MSE = MSE(y_test, y_pred_test)
test_RMSE = RMSE(y_test, y_pred_test)
print("Mean Squared Error:", test_MSE)
print("Root Mean Squared Error:", test_RMSE)
```

```
Mean Squared Error: 1.3750309955795914
Root Mean Squared Error: 1.172617156440921
```

In [ ]:

In [60]:

```python
# Plot training, validation, and test data with predictions
plt.figure(figsize=(12, 6))

# Plot training data and predictions
plt.plot(train_y.index, train_y, label="Train Actual", color="blue", linestyle="-")
plt.plot(train_y.index, y_pred_train, label="Train Prediction", color="blue", linestyle="--")

# Plot validation data and predictions
plt.plot(validation_y.index, validation_y, label="Validation Actual", color="green", linestyle="-")
plt.plot(validation_y.index, y_pred_validation, label="Validation Prediction", color="green", linestyle="--")

# Plot test data and predictions
plt.plot(test_y.index, test_y, label="Test Actual", color="red", linestyle="-")
plt.plot(test_y.index, y_pred_test, label="Test Prediction", color="red", linestyle="--")

# Formatting the plot
plt.title("Actual vs Predicted Values")
plt.xlabel("Index")
plt.ylabel("Values")
plt.legend()
plt.grid(True)

plt.show()
```



In [47]:

```python
#Visualizing Least Squares Estimation on a Toy Dataset

toy_data = pd.read_csv("toy_data.csv")
toy_data.head(10)
```

Out[47]:

| | X1 | X2 | Y |
|---|---|---|---|
| **0** | 5.530492 | 8.136530 | 53.470131 |
| **1** | 5.111720 | 0.846906 | 15.925409 |
| **2** | 9.011047 | 6.510469 | 54.649639 |
| **3** | 7.806497 | 0.349096 | 24.003095 |
| **4** | 2.047190 | 1.057417 | 14.739897 |
| **5** | 3.640398 | 8.856639 | 55.696555 |
| **6** | 0.973068 | 4.879262 | 28.903875 |
| **7** | 6.321576 | 4.464032 | 35.481217 |
| **8** | 2.420427 | 7.237535 | 41.646911 |
| **9** | 5.451504 | 8.074263 | 53.899325 |

In [48]:

```python
toy_data.shape
```

Out[48]:

```
(100, 3)
```

In [49]:

```python
X = toy_data.iloc[:, : -1].values
y = toy_data.iloc[ : , -1:].values
```

In [50]:

```python
# Compute the estimated parameters using the LSE method.
toy_beta = LSE(X, y)
print("Estimated parameters:", toy_beta)
```

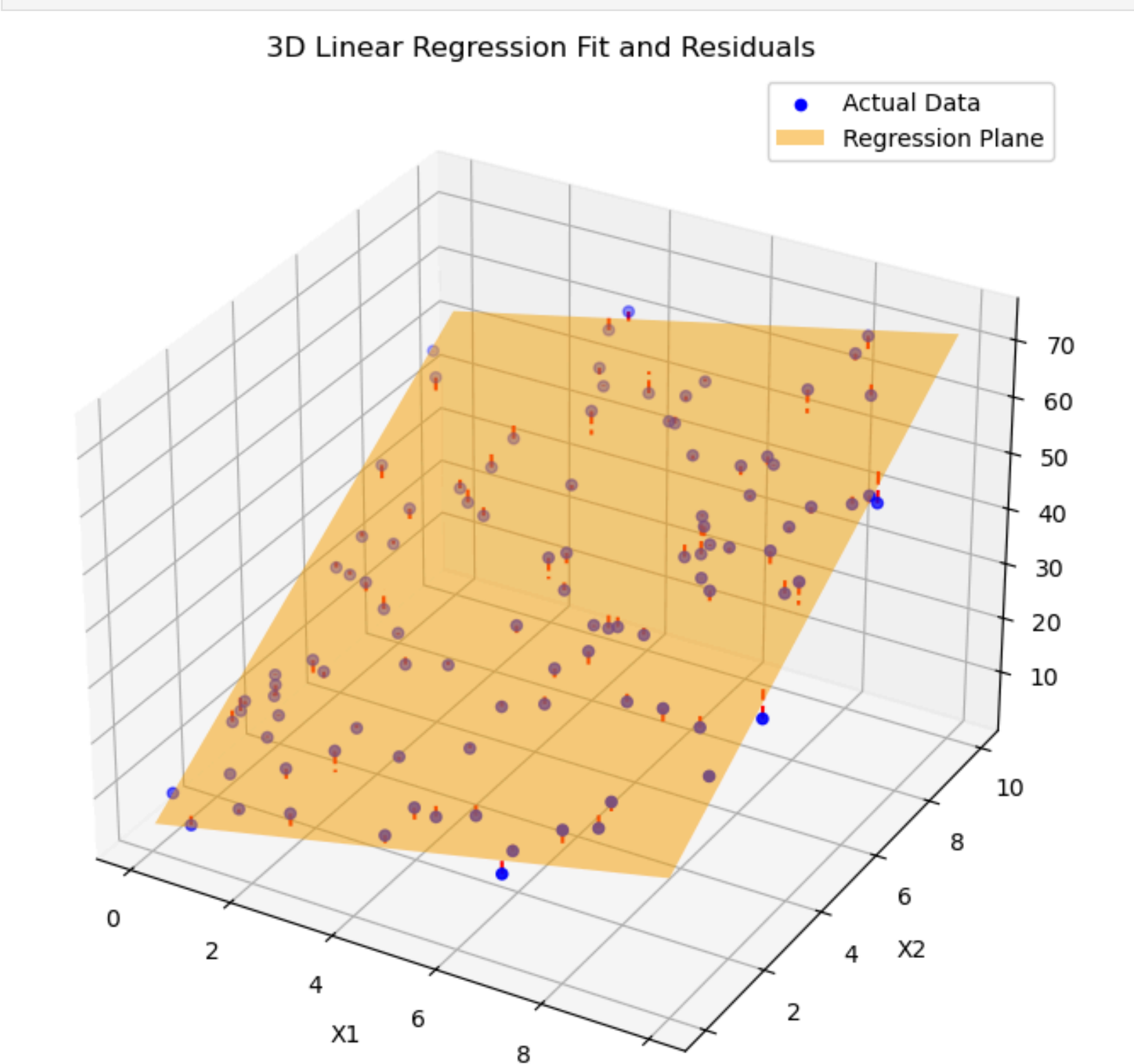
```
Estimated parameters: [[3.66345247]
 [1.96836519]
 [4.93212246]]
```

In [51]:

```python
y_hat = predic_function(X, toy_beta)
```

In [52]:

```python
from mpl_toolkits.mplot3d import Axes3D

# Create a 3D plot
fig = plt.figure(figsize=(10, 8))
ax = fig.add_subplot(111, projection='3d')

# Scatter plot of actual data points
scat = ax.scatter(toy_data['X1'], toy_data['X2'], toy_data['Y'], color='blue', label='Actual Data')

# Create a meshgrid for the regression plane
x1_range = np.linspace(toy_data['X1'].min(), toy_data['X1'].max(), 10)
x2_range = np.linspace(toy_data['X2'].min(), toy_data['X2'].max(), 10)
x1_mesh, x2_mesh = np.meshgrid(x1_range, x2_range)
y_mesh = toy_beta[0] + toy_beta[1] * x1_mesh + toy_beta[2] * x2_mesh  # Regression plane

# Plot the regression plane
plane = ax.plot_surface(x1_mesh, x2_mesh, y_mesh, color='orange', alpha=0.5, rstride=100, cstride=100)

# Plot residuals
for i in range(len(y)):
    ax.plot([X[i, 0], X[i, 0]], [X[i, 1], X[i, 1]], [y[i, 0], y_hat[i, 0]], 'r--', label='Residuals' if i == 0 else "")

# Labels and legend
ax.set_xlabel('X1')
ax.set_ylabel('X2')
ax.set_zlabel('Y')
ax.legend([scat, plane], ['Actual Data', 'Regression Plane'], loc='best')
ax.set_title('3D Linear Regression Fit and Residuals')

plt.show()
```



In [ ]:

In [ ]:

In [ ]: